



LAB 1- LAN

- Switch Basic Operation
- Learning outcomes:
 - CONNECT AND CONFIGURE SWITCH IN LOCAL AREA NETWORK (P4)
 - DIFFERENTIATE THE HARDWARE USED IN LAN (C4)

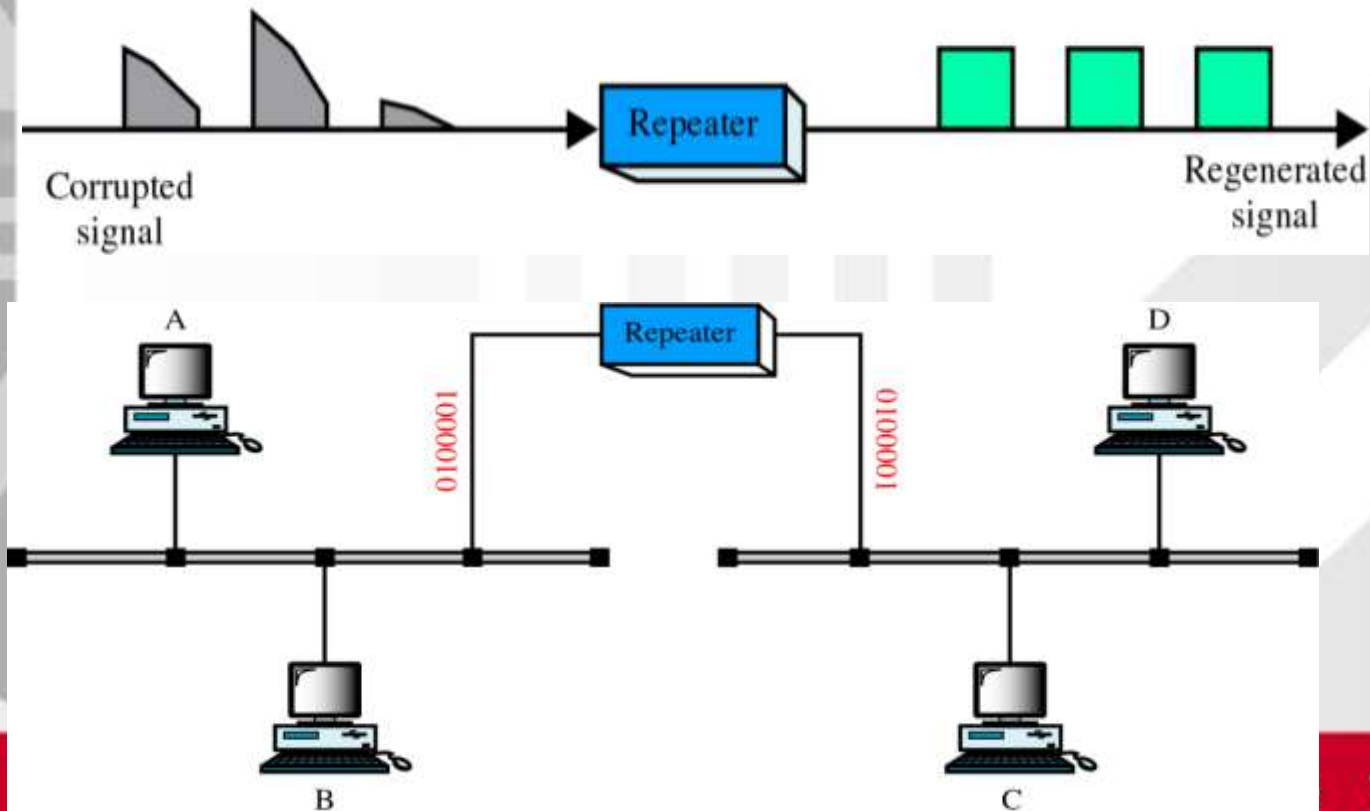


Connecting LAN

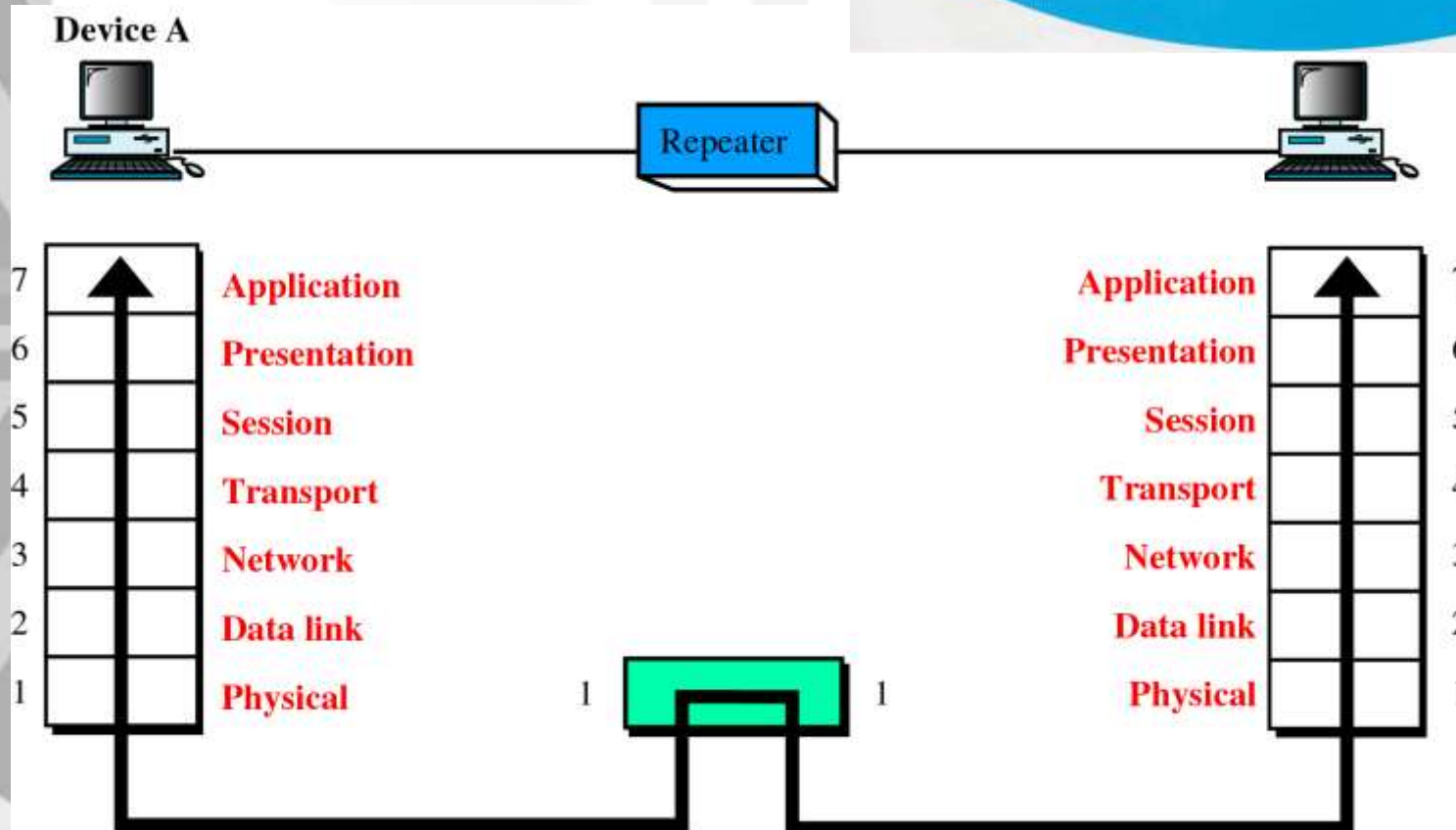
- Hubs/Repeaters
- Switches
- Bridges
- Routers
- Gateways

Hubs/Repeaters

- regenerate electrical data signals
- takes the incoming packet and copies it out to all other ports
- operate at the Physical Layer

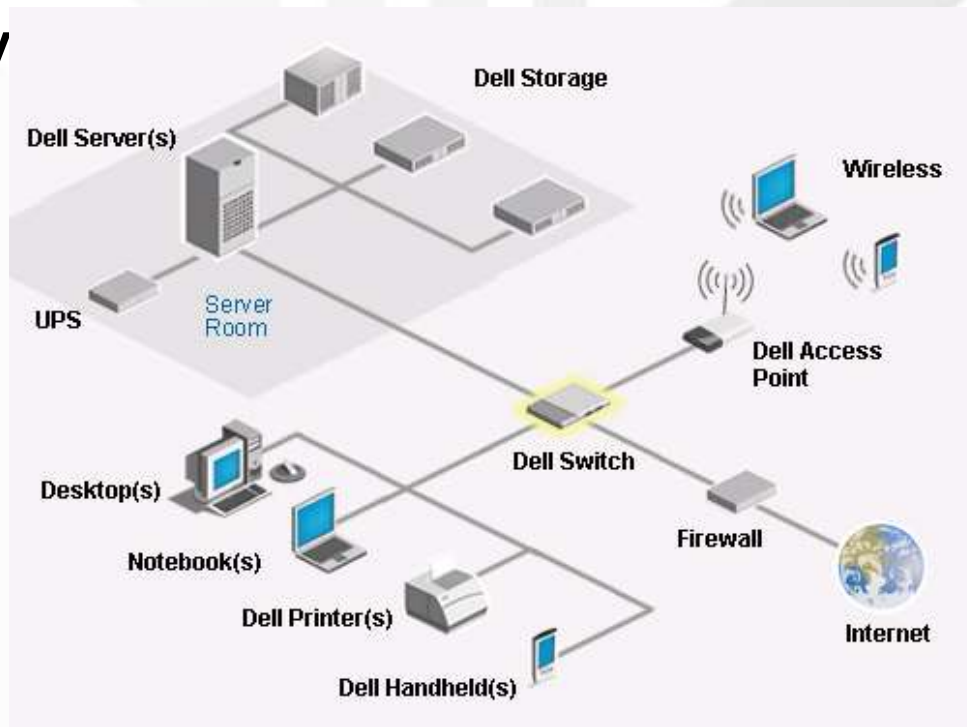


Hubs/Repeaters



Switches

- route data to the destination address only (based on physical address)
- isolate data flow based on connection
- typically



Switches

*Access Switch
Seminar Room
(Block A :Aras 1)*



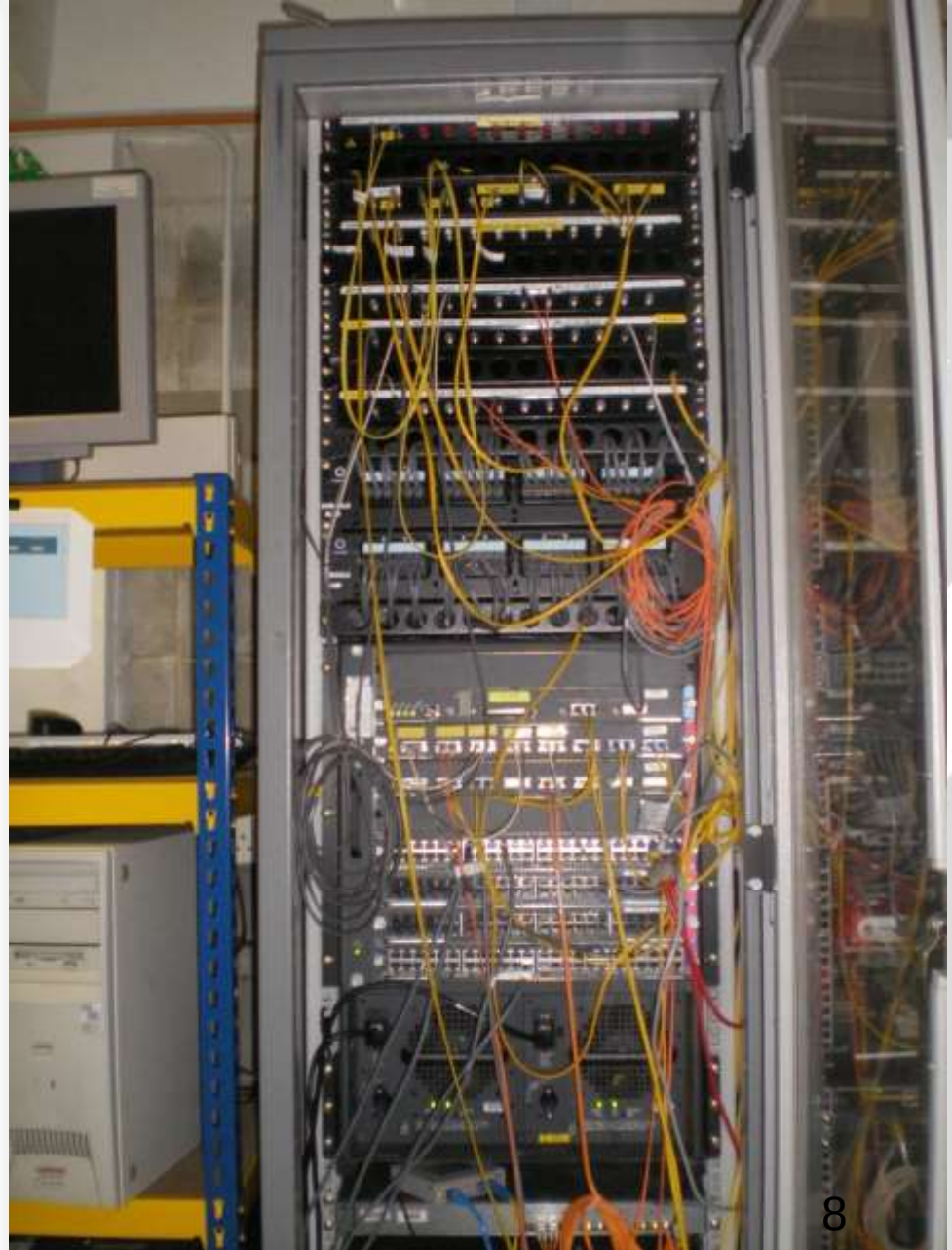
Switches

*Access Switch
PDR Room
(Block A, Aras 3)*



Switches

*Core Switches
Server Room
(Block A, Aras 1)*

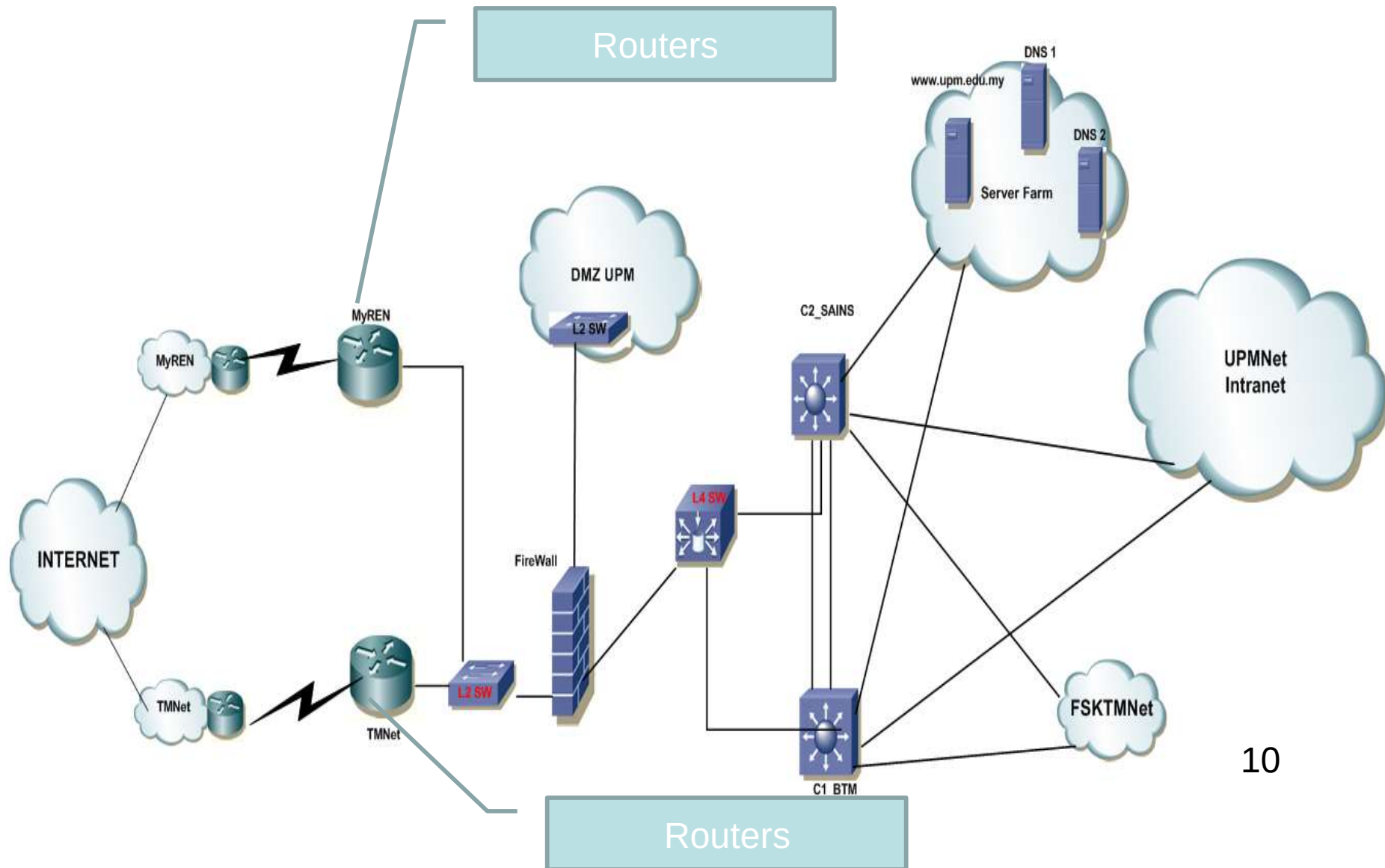




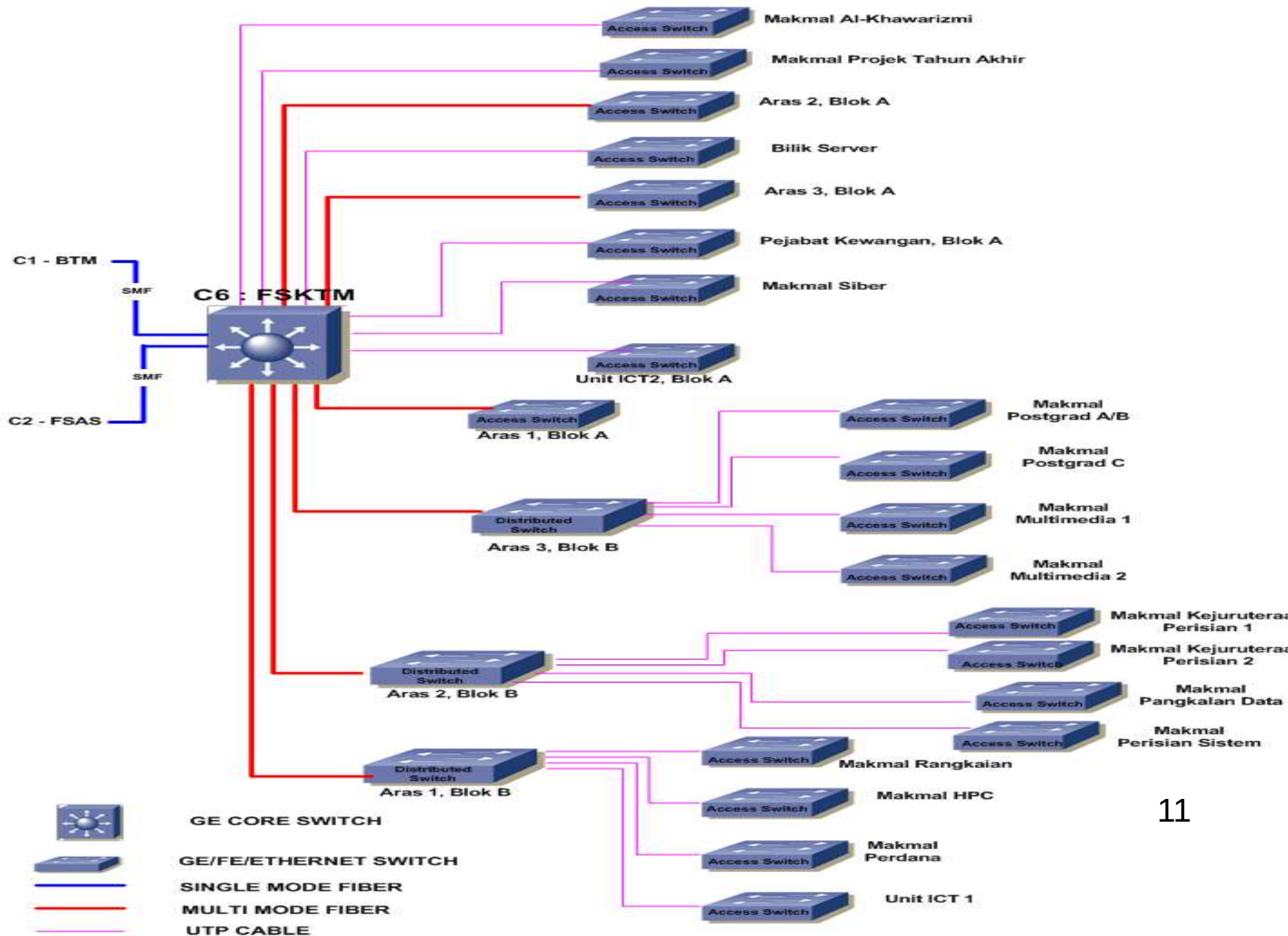
FSKTMNet

- Started operation : Jun 2001
- Number of nodes: 1284
- Network Equipment:
 - 1 Core switch
 - 3 Distributed switch
 - 34 Access switch

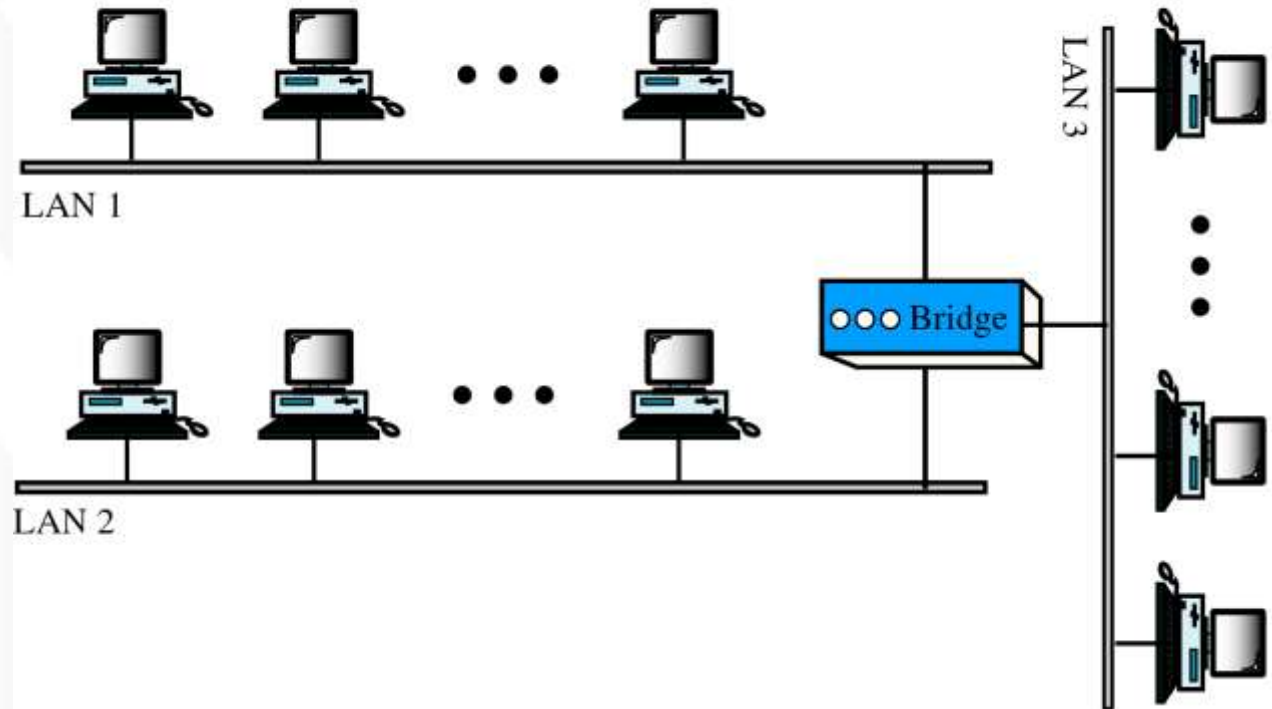
UPM Network Diagram



FSKTMNET



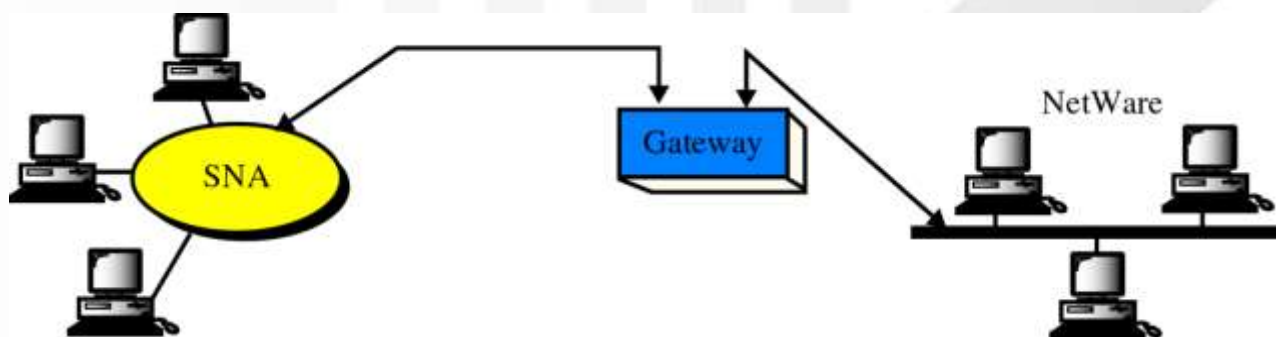
Bridges



- work at Link Layer
- used to connect 2 (LANs) or 2 segments of the same LAN
- build up a table of which port a machine is connected to
- filter data packets based on the MAC address

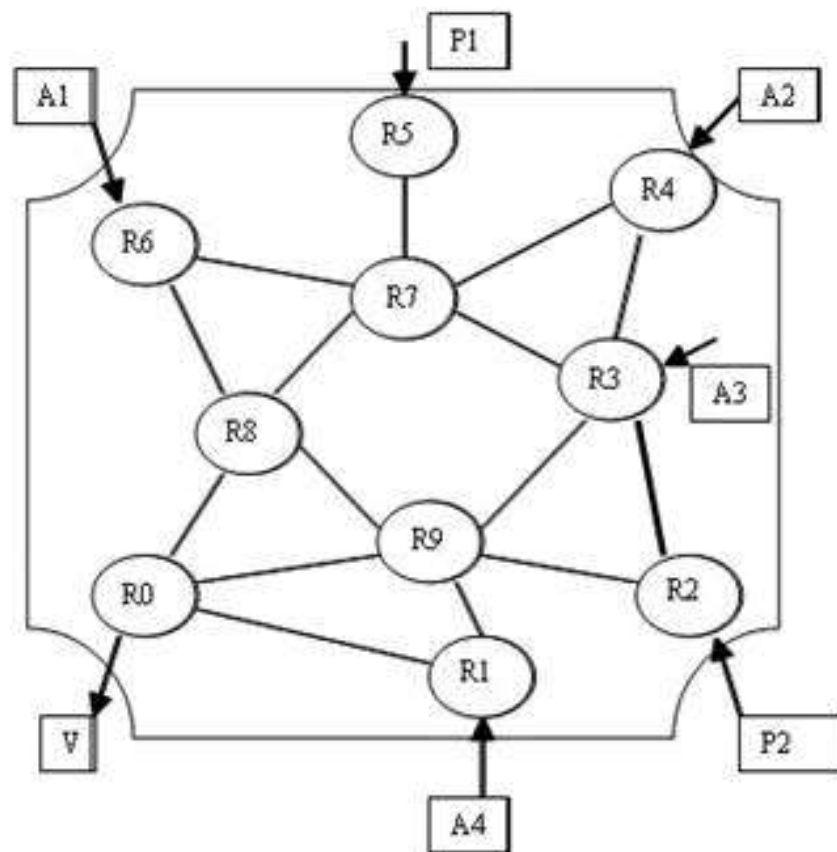
Gateways

- also called **protocol converters**
- can operate at any layer of the OSI Model
- can be rate converters, fault isolators signal translators etc as to provide system interoperability
- a gateway is an essential feature of most routers, although other devices can function as a gateway
- an entry point and an exit point in a network



Routers

- forward data depending on the IP address
- work in networks using similar protocols
- use routing algorithms to get the best path
- operate in the 1, 2 and 3 layers



tracert is.saga-u.ac.jp



Routers

```
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Administrator>tracert is.saga-u.ac.jp

Tracing route to is.saga-u.ac.jp [133.49.200.1]
over a maximum of 30 hops:

  0  <1 ms    <1 ms    <1 ms    172.16.60.252
  1  <1 ms    <1 ms    <1 ms    192.168.0.33
  2  <1 ms    <1 ms    <1 ms    192.168.1.251
  3  544 ms   535 ms   537 ms   119.40.118.254
  4  542 ms   535 ms   543 ms   119.40.118.1
  5  535 ms   527 ms   542 ms   202.171.193.61
  6  557 ms   562 ms   555 ms   202.171.192.81
  7  582 ms   588 ms   591 ms   202.171.192.82
  8  588 ms   582 ms   587 ms   202.179.249.69
  9  693 ms   677 ms   671 ms   202.179.249.50
 10  649 ms   633 ms   631 ms   tokyo1-dc-RM-GE-1-2-0-107.sinet.ad.jp [150.99.19
8.201]
 11  667 ms   667 ms   *        nagoya-dc-RM-AE-0-11.sinet.ad.jp [150.99.203.261]
 12  *        *        *        Request timed out.
 13  *        *        *        Request timed out.
 14  *        *        741 ms   fukuoka-dc-RM-GE-6-1-0-11.sinet.ad.jp [150.99.20
3.46]
 15  734 ms   743 ms   738 ms   saga-u.gw.sinet.ad.jp [150.99.208.18]
 16  748 ms   768 ms   769 ms   sadaigw1.cc.saga-u.ac.jp [133.49.200.250]
 17  767 ms   776 ms   774 ms   133.49.200.254
 18  *        *        *        Request timed out.
 19  *        *        *        Request timed out.
 20  *        *        *        Request timed out.
 21  *        *        *        Request timed out.
 22  *
```

route print

```
C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Administrator>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...00 21 9b 6c e6 77 ..... Intel(R) 82566DM-2 Gigabit Network Connection -
Packet Scheduler Miniport
=====

Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          172.16.61.254    172.16.61.168     20
127.0.0.0                  255.0.0.0        127.0.0.1        127.0.0.1         1
172.16.61.0                255.255.255.0    172.16.61.168    172.16.61.168     20
172.16.61.168              255.255.255.255  127.0.0.1        127.0.0.1         20
172.16.255.255             255.255.255.255  172.16.61.168    172.16.61.168     20
224.0.0.0                  240.0.0.0        172.16.61.168    172.16.61.168     20
255.255.255.255           255.255.255.255  172.16.61.168    172.16.61.168     1
Default Gateway:          172.16.61.254
=====

Persistent Routes:
None

C:\Documents and Settings\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

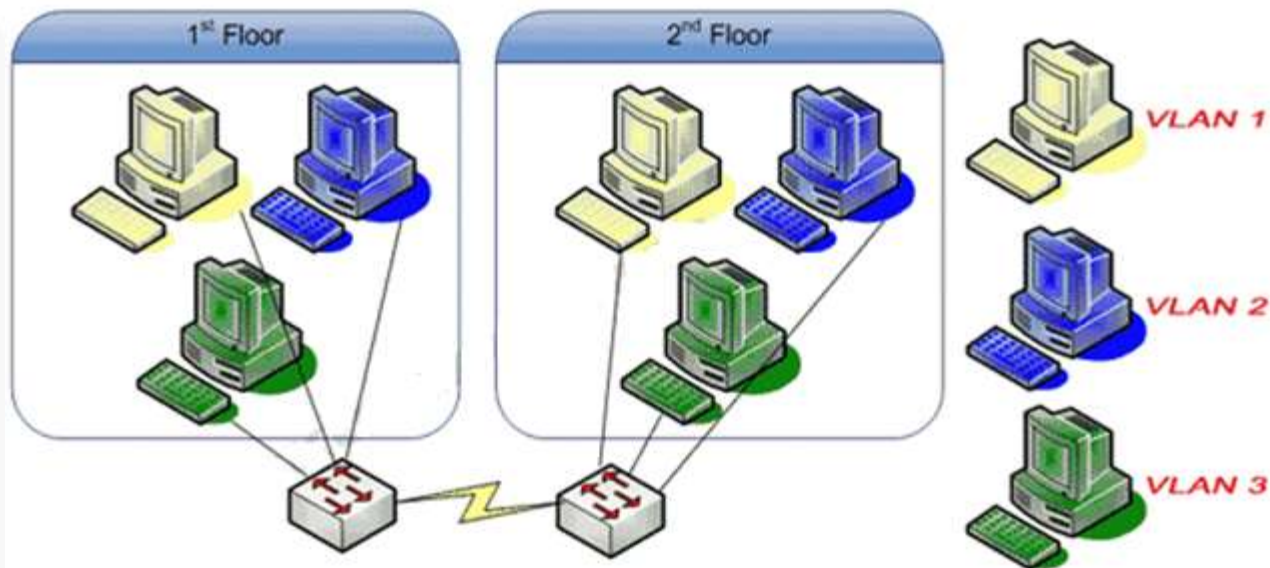
    Connection-specific DNS Suffix  . : upm.edu.my
    IP Address. . . . .               : 172.16.61.168
    Subnet Mask . . . . .             : 255.255.255.0
    Default Gateway . . . . .         : 172.16.61.254

C:\Documents and Settings\Administrator>
```

VLAN...

A VLAN (virtual local area network) is a collection of nodes that are grouped together in a single broadcast domain that is based on something other than physical location.

In simpler term, VLAN is a logical groups of devices that are defined by a software configuration, usually in switches.





Why VLAN?

Security - Separating systems that have sensitive data from the rest of the network decreases the chances that people will gain access to information they are not authorized to see.

Departments/Specific job types - Companies may want VLANs set up for departments that are heavy network users (such as multimedia or engineering), or a VLAN across departments that is dedicated to specific types of employees (such as managers or sales people).

Performance/Bandwidth - Careful monitoring of network use allows the network administrator to create VLANs that reduce the number of router hops and increase the apparent bandwidth for network users.

Broadcasts/Traffic flow - Since a principle element of a VLAN is the fact that it does not pass broadcast traffic to nodes that are not part of the VLAN, it automatically reduces broadcasts.



Types of VLAN

[1] Port-Based VLAN

- Also known as Static VLAN
- **Specified by switch port.** Each physical switch port is configured with an access list specifying membership in a set of VLANs.
- Example:
 - A 12 port fast Ethernet switch is split for the creation of 2 VLANs.
 - The first 6 ports are associated with VLAN1 and the last 6 ports are associated with VLAN2.
 - If a machine is moved from port 3 to port 11, it will effectively change VLANs.



Types of VLAN

[2] MAC-Based VLAN

- Also known as Dynamic VLAN
- **Specified by MAC address.** A switch is configured with an access list mapping individual MAC addresses to VLAN membership.
- Example:
 - A sys admin will enter MAC addresses for all machines connecting to the switch.
 - These addresses will be stored in a memory chip inside the switch that forms a database of local MAC addresses.
 - Each MAC address can then be associated with a certain VLAN.
 - If a machine is moved, it will retain the original VLAN membership regardless of its port number.



Types of VLAN

[3] Protocol-Based VLAN

- A switch is configured with a list of mapping layer 3 protocol types to VLAN membership - thereby filtering IP traffic from nearby end-stations using a particular protocol such as IPX.

[4] ATM VLAN

- using LAN Emulation (LANE) protocol to map Ethernet packets into ATM cells and deliver them to their destination by converting an Ethernet MAC address into an ATM address.

Sample VLAN Configuration:

Splitting VLAN per switch port

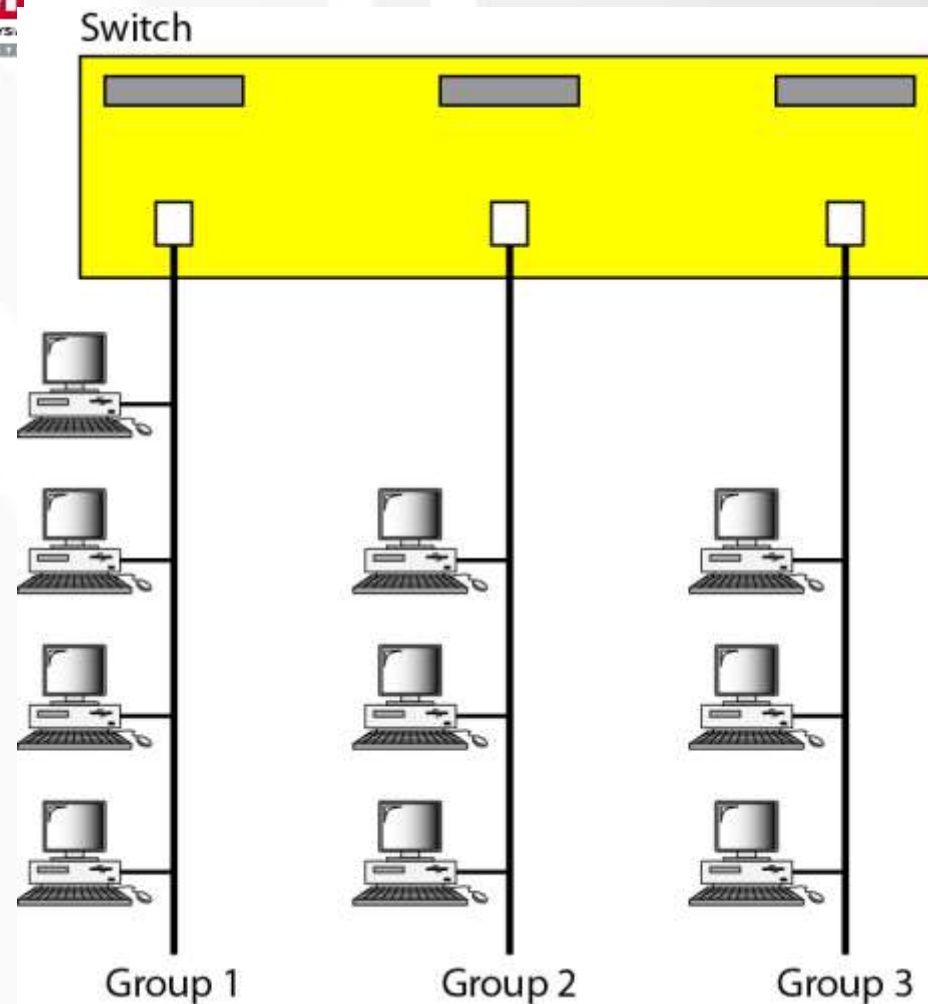
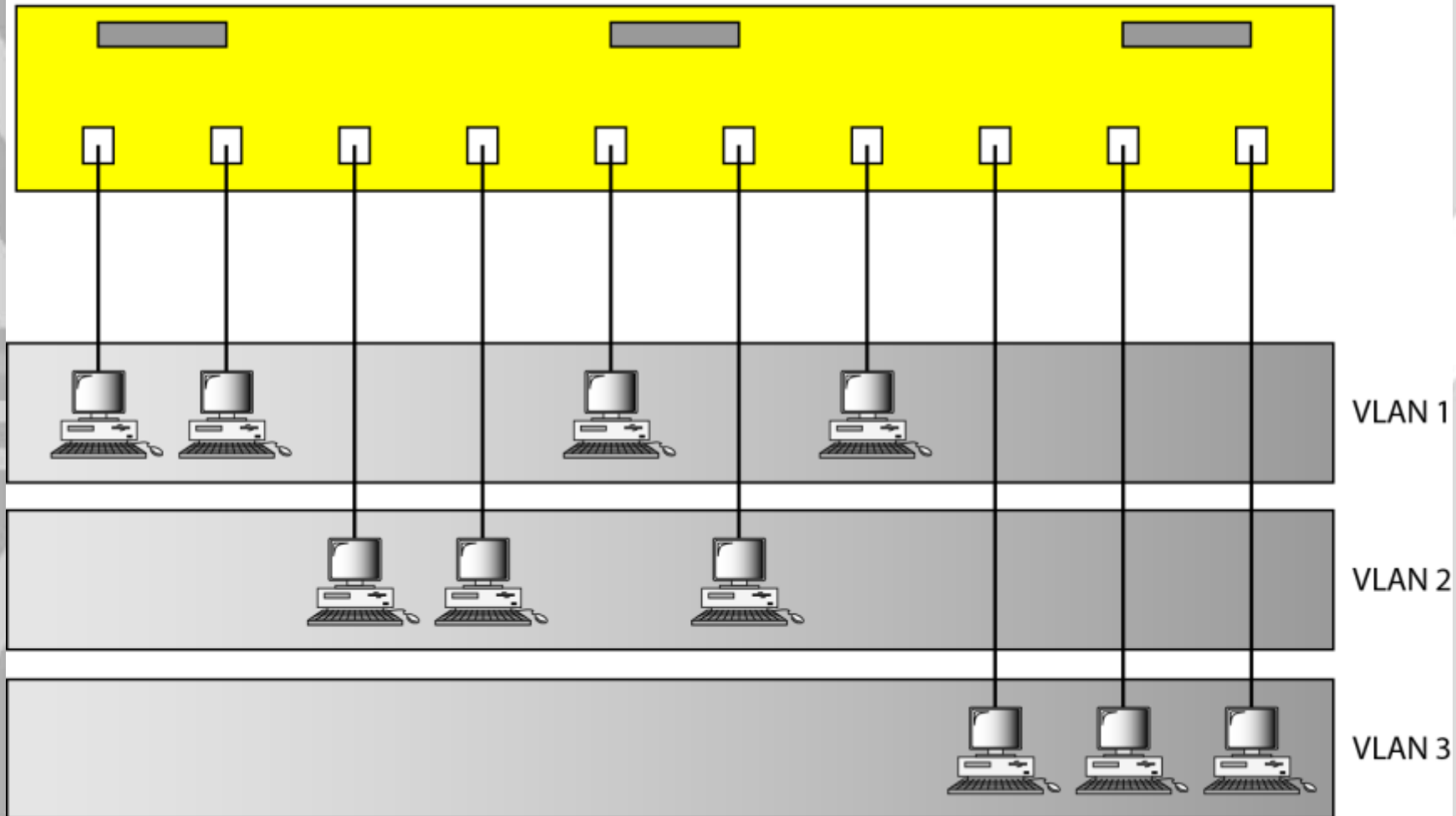


Figure 15.15 *A switch connecting three LANs*

Sample VLAN Configuration:

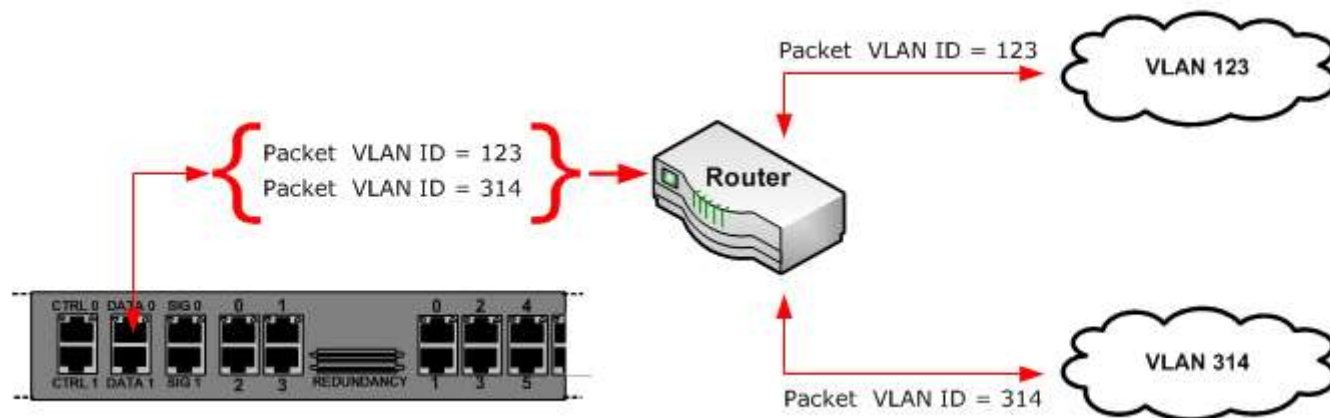
Specified VLAN per switch port

Switch with VLAN software



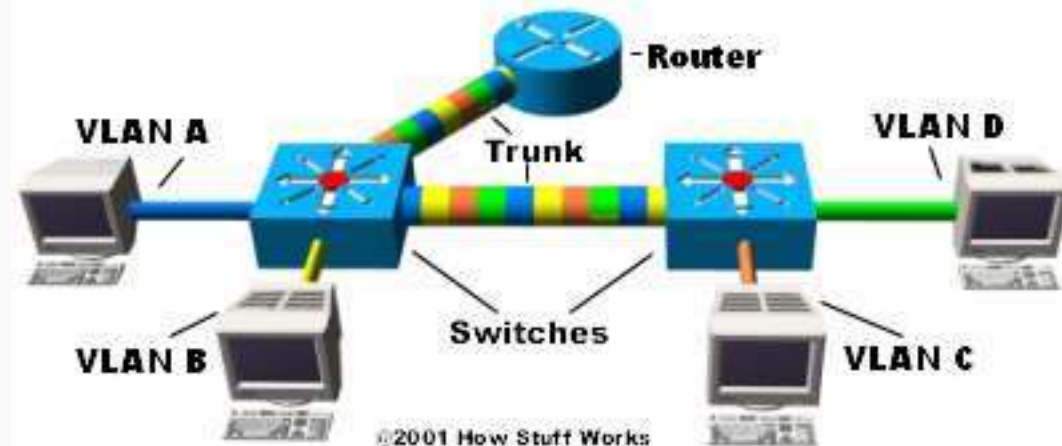
VLAN Tagging

When VLANs span multiple switches, VLAN Tagging is required. VLAN Tagging is the practice of inserting a VLAN ID into a packet header in order to identify which VLAN the packet belongs to. More specifically, switches use the VLAN ID to determine which port(s), or interface(s), to send a broadcast packet to.

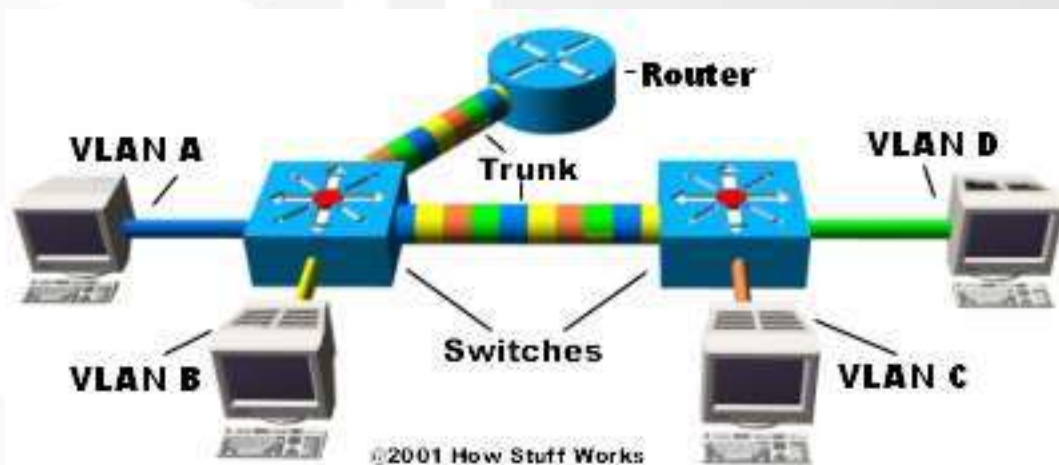


VLAN Trunking

When VLANs span multiple switches, a trunk data link is required between the switches. This is called VLAN Trunking and VLAN trunking protocol (VTP) is the protocol that switches use to communicate among themselves about VLAN configuration that are which port(s), or interface(s), to send a broadcast packet

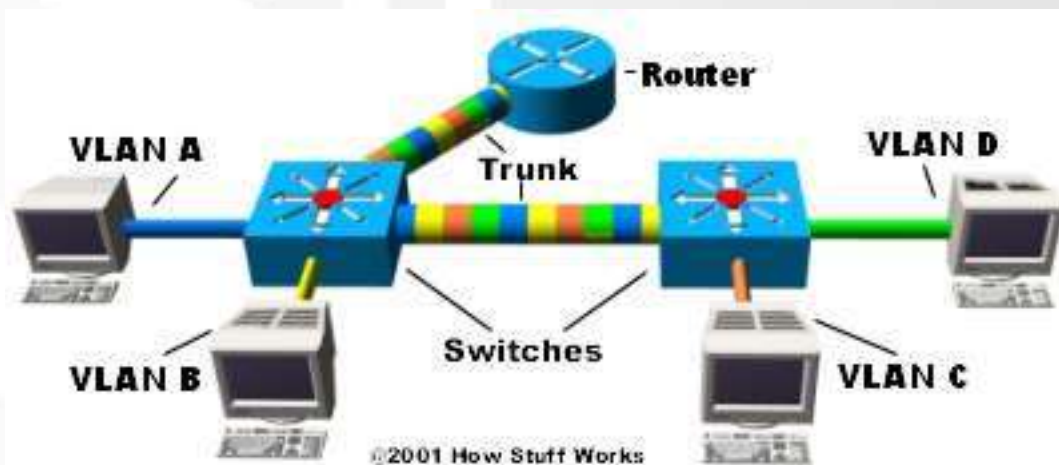


VLAN Trunking: Example Scenario



- Each switch has two VLANs.
- On the 1st switch, VLAN A and VLAN B are sent through a single port (trunked) to the router and through another port to the second switch.
- VLAN C and VLAN D are trunked from the second switch to the first switch, and through the first switch to the router. This trunk can carry traffic from all four VLANs.
- The trunk link from the first switch to the router can also carry all four VLANs.

VLAN Trunking: Example Scenario



- Using the router, the VLANs can communicate with each other via the trunking connection between the two switches
- Data from a computer on VLAN A that needs to get to a computer on VLAN B (or VLAN C or VLAN D) must travel from the switch to the router and back again to the switch.
- Because of the transparent bridging algorithm and trunking, both PCs and the router think that they are on the same physical segment!